DATA STREAM MINING • NOVA IMS

Practical Assignment I

Business-Oriented Data Stream Mining

From Real-Time Weather Data to Operational Decisions

Core idea
Use a real weather stream to define an operational event, compare two Portuguese cities, evaluate incremental classifiers correctly, and convert evidence into a business recommendation and a technical recommendation.

Practical Assignment Handbook

Weeks 1–6 integration | Pair work | Technical report + audit package

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CHAPTER 1

Assignment Overview

Central principle

The model does not make the business decision. You do.

Practical Assignment I consolidates the first six weeks of the course into one applied, business-oriented Data Stream Mining study. Each pair works with two Portuguese cities and one operational event target derived from Open-Meteo weather data. Your task is to understand the two streams, transform the business context into a leakage-free prediction problem, evaluate incremental learners under a correct chronological protocol, and use evidence to support two distinct recommendations.

Business recommendation: Which city appears more suitable from the *weather-related operational perspective* defined in your assignment card?

Technical recommendation: Which streaming approach is most appropriate for continuous event prediction or monitoring, considering predictive behaviour, event detection, computational cost, and practical constraints?

Each pair receives one unique assignment code, **PAxx**, defining a business scenario, City A, City B, one official binary target (T1–T6), and one guiding business question. The full allocation catalogue appears in Chapter 19.

1.1 What this assignment is - and is not

This is **not** a contest to obtain the highest Accuracy. It is a structured investigation of a real stream. You must connect data characteristics, temporal behaviour, feature construction, predictive evidence, computational evidence, and business interpretation.

The weather analysis supports only the operational dimension explicitly stated in your assignment card. A real location decision could also depend on rent, wages, taxes, transport infrastructure, regulation, customer access, or other factors that are outside this dataset.

CHAPTER 2

Learning Objectives

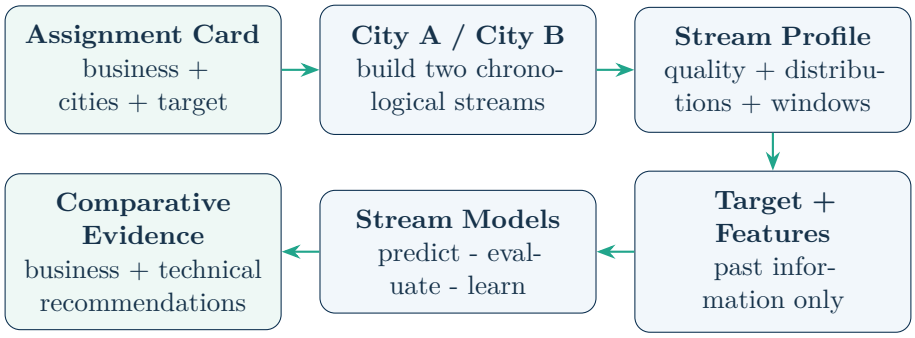
By completing the assignment, you should be able to:

- organise and inspect a real chronological data stream;
- construct a binary operational target without using future information;
- distinguish data quality problems from valid extreme observations;

- build lagged and window-based stream features;
- evaluate incremental classifiers using a prequential protocol;
- interpret Accuracy, Macro-F1, Precision, Recall, and the confusion matrix in context;
- compare single incremental learners with streaming ensembles;
- measure predictive and update cost per event;
- separate a business recommendation from a technical model recommendation;
- document a reproducible, auditable experiment in a concise technical report.

CHAPTER 3

How the Assignment Works



3.1 A common protocol, different evidence

For a fair comparison, both cities must use the same target definition, threshold policy, feature logic, warm-up rule, chronological protocol, and evaluation metrics. Models compared within a city must receive the same chronological observations.

Core streaming rule

PREDICT → EVALUATE → LEARN

The current label must never influence the prediction that is being scored for that same event.

3.2 Original contribution

Each pair must create **at least two analyses or visualisations that are not direct reproductions of figures supplied in the course starter code**. Appropriate examples include event frequency over time, rolling event rate, distribution comparisons, hour-of-day profiles, consecutive adverse-event duration, feature correlations, window statistics, class-imbalance evolution, and cost-performance plots.

Every figure must have a job

Do not include a plot simply because the code generated it. Every figure should support a question, observation, comparison, or recommendation in the report.

CHAPTER 4

Pair Responsibilities and Individual Accountability

Responsibility	Student A	Student B
Primary city	City A	City B
Data quality	Inspect and document City A	Inspect and document City B
Stream profile	Statistics, temporal evolution, windows	Statistics, temporal evolution, windows
Target	Construct and inspect target frequency	Construct and inspect target frequency
Models	Run and understand City A outputs	Run and understand City B outputs
Interpretation	Explain evidence for City A	Explain evidence for City B

Joint responsibility includes the business framing, common target protocol, feature-engineering choices, model comparison, comparative city analysis, business recommendation, technical recommendation, limitations, and final integration of the report.

Individual responsibility means that each student has a primary city to develop and understand. The final deliverable must still read as one coherent technical report in which the two streams are compared under a common experimental design.

The submission receives a group grade, but individual grades may differ when there is clear evidence of unequal participation or inability to explain the submitted work. Either student may be asked to explain any submitted code, figure, table, metric, transformation, or interpretation.

CHAPTER 5

From Weather Data to a Business Event

The raw stream contains weather variables such as temperature, relative humidity, precipitation, and wind speed. Your assignment converts these measurements into an operational event that a business may want to anticipate.

For an event observed at time t, construct the prediction problem as

$$X_t = \text{information available before time } t, \qquad y_t = \text{event observed at time } t.$$

The assignment therefore asks two different kinds of questions:

1. **Descriptive/business:** How often, how persistently, and under what temporal patterns does the event occur in each city?
2. **Predictive/technical:** How effectively and efficiently can incremental models predict that event from information available beforehand?

CHAPTER 6

The Six Official Targets

All groups must use the target specified by their assignment card. Targets T2–T5 use thresholds computed from the **pooled warm-up observations of both assigned cities**. The threshold is estimated once and then frozen before evaluation.

ID	Target	Definition	Operational interpretation
T1	Rain Event	$y_t = 1$ if $P_t \geq 0.1$ mm; otherwise 0.	Rain-related operational interruption.
T2	Heat Stress Event	$\tau_{T,hot} = Q_{0.80}(T)$ from pooled warm-up; $y_t = 1$ if $T_t \geq \tau_{T,hot}$.	Relatively high-temperature condition for the assigned city pair.
T3	Cold Stress Event	$\tau_{T,cold} = Q_{0.20}(T)$ from pooled warm-up; $y_t = 1$ if $T_t \leq \tau_{T,cold}$.	Relatively low-temperature condition for the assigned city pair.
T4	Humid vs Dry Regime	$\tau_H = Q_{0.50}(H)$ from pooled warm-up; $y_t = 1$ if $H_t \geq \tau_H$, else 0.	Humid regime versus a relatively drier regime.
T5	Strong Wind Event	$\tau_W = Q_{0.80}(W)$ from pooled warm-up; $y_t = 1$ if $W_t \geq \tau_W$.	Relatively strong-wind condition for the assigned pair.
T6	Adverse Outdoor Event	$y_t = 1$ if T1 is true **or** T5 is true; otherwise 0.	Rain-or-strong-wind condition that may disrupt outdoor operations.

Threshold rule - mandatory

Estimate thresholds from the warm-up. Freeze them. Never recompute them using future observations.

6.1 Why pooled warm-up thresholds?

Using the same threshold for the two cities makes their event frequencies directly comparable while keeping the rule independent from future evaluation observations. A pair-specific quantile also avoids imposing an arbitrary universal value when the purpose is to study relative operational stress in the assigned context.

CHAPTER 7

Avoiding Target Leakage

Target leakage occurs when information that would not have been available at prediction time is used as a predictor. In this assignment, the most direct form of leakage is using the current value that defines the current target.

Leakage example - Rain Event at time t

INVALID: use `precipitation(t)` to predict `Rain(t)`. The input already contains the answer.

VALID: use `precipitation(t-1)`, `humidity(t-1)`, `temperature(t-1)`, `wind(t-1)`, and rolling features computed exclusively from observations before t.

The same principle applies to all targets. If temperature defines Heat Stress at time t, the current temperature belongs to the target definition and must not appear in X_t. Lagged temperature and rolling statistics ending at $t-1$ are valid.

7.1 Temporal alignment checklist

Before training a model, verify that:

- all lagged predictors refer to observations strictly before the target time;
- rolling windows end at $t-1$ or earlier;
- target thresholds were estimated only from warm-up observations;
- scaling or other fitted preprocessing does not use future evaluation observations;
- chronology is preserved throughout replay and evaluation.

CHAPTER 8

Required Feature Representations

Every pair must evaluate two representations. Additional justified features are welcome if they remain leakage-free.

8.1 Representation A - Previous-State Features

At minimum:

- `temperature(t-1)`
- `relative_humidity(t-1)`
- `precipitation(t-1)`
- `wind_speed(t-1)`

8.2 Representation B - Stream-Aware Temporal Context

Representation B contains all of Representation A plus past-only temporal context. It must include at least:

- one lagged change/delta feature, e.g. $\Delta T_t = T_{t-1} - T_{t-2}$;
- one rolling mean, e.g. $\bar{T}_t^{(6)} = \frac{1}{6}\sum_{i=1}^{6} T_{t-i}$;
- one rolling standard deviation;
- one recent event-frequency or recent-condition feature, e.g. rainy hours in the previous 24 hours.

Recommended window sizes are 6 hours and/or 24 hours. You may justify alternatives when they better match the operational scenario.

Common comparison question

Does recent temporal context improve event prediction compared with using only the most recent observation?

CHAPTER 9

Required Stream Characterisation

Before modelling, each student must understand the stream assigned to their city. At minimum, the City A and City B analyses must provide evidence on:

1. data quality, missing values, and invalid observations;

2. descriptive statistics for the main weather variables;
3. temporal evolution of relevant variables;
4. target frequency and class distribution;
5. rolling/window behaviour;
6. unusual or extreme observations;
7. at least one interpretation connected to the business context.

Extreme does not mean erroneous

Do not automatically remove extreme weather values. An unusually high temperature, strong-wind observation, or heavy-rain episode may be exactly the operational event the business needs to detect. Remove or correct observations only when you have evidence that they are invalid, corrupted, or inconsistent with the data definition. Document any such intervention.

9.1 Useful descriptive evidence

Appropriate evidence may include distributions, summary tables, temporal plots, rolling mean/variance, event-rate evolution, consecutive-event duration, and hour-of-day profiles. Choose analyses that help explain the stream and the operational problem rather than maximising the number of plots.

CHAPTER 10

Required Models

The following five learners are mandatory for both cities and both required feature representations:

	Model	Why it is included
1	Gaussian Naive Bayes	Incremental probabilistic baseline with compact sufficient statistics.
2	Online Logistic Regression	Incremental linear learner whose parameters evolve continuously.
3	Hoeffding Tree	Incremental tree that can evolve its structure from stream evidence.
4	Online Bagging	Ensemble that converts bootstrap logic into online resampling.
5	Adaptive Random Forest	Adaptive ensemble of randomised stream trees.

10.1 Optional extensions

You may add Leveraging Bagging, Streaming Random Patches, the public ENFS-Uni0 implementation, or another justified incremental learner available in the course ecosystem. Optional models do not replace the five mandatory models.

All compared models must receive the same chronological observations under the same protocol for a given city and feature representation.

CHAPTER 11

Evaluation Metrics

11.1 Required predictive evidence

Report:

- Accuracy;
- Macro-F1;
- Precision for the positive/event class;
- Recall for the positive/event class;
- Confusion Matrix.

A high Accuracy value is not sufficient evidence

When the operational event is rare, a classifier can obtain high Accuracy by predicting the majority class most of the time while still missing the events that matter. Inspect class balance, Precision, Recall, Macro-F1, and the confusion matrix before drawing conclusions.

11.2 Required computational evidence

Report:

- prediction time per event;
- update time per event.

Where meaningful and available, model or population complexity may be reported as supporting evidence (for example tree nodes or ensemble size), but a common structural complexity measure is not mandatory across all algorithms.

CHAPTER 12

Experimental Rules

A valid experiment must satisfy the following controls:

1. preserve chronological order; do not use a random train/test split;
2. use the common warm-up policy for the two cities;

3. estimate data-derived thresholds from pooled warm-up data only;
4. freeze thresholds before the evaluation stream begins;
5. construct all predictors from information available before the target event;
6. evaluate before learning the current label;
7. use fresh model instances when repeating an experiment or changing a representation;
8. compare models on the same evaluation history;
9. record relevant model parameters, package versions, random seed, and historical period;
10. change one scientific factor at a time when interpreting a controlled comparison.

Fairness rule

When you compare models, representations, or cities, identify clearly what is being held constant and what is being changed. A comparison is only meaningful when the protocol is controlled.

CHAPTER 13

Business Interpretation

The report must end with two separate recommendations.

13.1 Business Recommendation

Answer the question on your assignment card: Which city appears more suitable **from the weather-related operational perspective considered in this study**, and what evidence supports that conclusion?

Use descriptive and temporal evidence such as event frequency, persistence, variability, rolling behaviour, and the operational meaning of those patterns. Predictive performance may support your understanding of the stream, but it does not automatically determine city suitability.

13.2 Technical Recommendation

Recommend a streaming approach for continuous prediction or monitoring in the specific case. Consider event detection, class imbalance, predictive metrics, temporal behaviour, prediction cost, update cost, and practical implementation constraints.

Predictive performance does not automatically determine business suitability.

13.3 Interpret evidence - do not narrate numbers

A strong discussion connects multiple forms of evidence. For example, a model may achieve a small Macro-F1 improvement while doubling update cost, or high Accuracy may coexist with poor Recall for the event class. Explain why those relationships matter for the operational context.

CHAPTER 14

Technical Report Structure

The recommended report length is **6–8 pages**, excluding references and appendices. The report must be concise, coherent, and evidence-driven.

No.	Section	Purpose
1	Business Context and Objective	State the decision, cities, target, and operational question.
2	Data and Stream Construction	Describe source, period, variables, chronology, warm-up, target, and preprocessing.
3	City A Analysis	Characterise City A and explain the evidence relevant to the business.
4	City B Analysis	Characterise City B using the same analytical logic.
5	Comparative Data Analysis	Compare the cities before discussing model performance.
6	Data Stream Mining Experiment	Define representations, models, parameters, protocol, and metrics.
7	Results and Discussion	Interpret predictive and computational evidence; do not only rank scores.
8	Business and Technical Recommendations	Provide two clearly separated recommendations.
9	Limitations	Explain what the data and experimental design cannot support.
10	References	Cite data, libraries, methods, and other sources used.
11	AI Assistance Statement	Disclose generative-AI assistance using the policy in Chapter 16.

Writing rule

Every figure must support an argument. A good technical report selects evidence; it does not paste every output generated by the script.

CHAPTER 15

Submission Package

Submit two Moodle deliverables:

1. `PAxx\_GroupYY\_Report.pdf`
2. `PAxx\_GroupYY\_Audit.zip`

15.1 Suggested audit ZIP structure

```
PAxx_GroupYY/
  README.md
  src/
    practical_assignment.py   (or notebook + supporting modules)
  data/
    raw snapshots or retrieval instructions
  results/
    metrics.csv
    configurations.csv
    figures/
  requirements.txt
```

The `README.md` must state the assignment code, student names, City A and City B, business scenario, target, historical period, Python version, package versions, random seed, and instructions to reproduce the main experiment.

Auditability

The instructor must be able to understand how every reported result was produced from the submitted code, configuration, and stored or retrievable data.

15.2 Reproducibility expectations

Use a fixed random seed where algorithms are stochastic, document package versions and model parameters, preserve chronology, record the historical period, and save or describe the data snapshots used. Reported figures and tables must correspond to the submitted experiment.

CHAPTER 16

Generative AI and LLM Policy

Generative AI tools may support programming work, but they may not replace the students' scientific interpretation.

Permitted assistance	Not permitted
Python syntax; debugging; library/API questions; refactoring; plotting syntax/code; code comments/documentation; understanding error messages.	Interpretation of experimental results; explanation of observed patterns; Results discussion; Business Recommendation; Technical Recommendation; final evidence-based conclusions.

Students remain responsible for understanding every submitted line of code, figure, table, and statement. Either student may be asked to explain any element of the work.

16.1 Mandatory AI Assistance Statement

Include one statement at the end of the report. Adapt it truthfully.

> *Generative AI tools were used to assist with Python debugging, plotting syntax and code organisation. They were not used to interpret experimental results or produce the analysis, discussion, business recommendation, technical recommendation, or evidence-based conclusions.*

> *No generative AI tools were used in the preparation of this assignment.*

CHAPTER 17

Assessment Rubric

The assignment is graded on the quality of the investigation, not on achieving a particular metric value.

Component	Weight	Strong evidence	Weak evidence
Business understanding & problem formulation	10%	Clear operational decision, correct interpretation of the assignment card, and scope aligned with available weather evidence.	Generic motivation, unclear decision, or claims beyond what the dataset can support.
Data preparation & stream construction	15%	Clean chronological stream, documented quality checks, correct warm-up and reproducible preprocessing.	Unclear chronology, undocumented cleaning, or inconsistent data construction.

Component	Weight	Strong evidence	Weak evidence
City-level data analysis	15%	Relevant statistics, temporal behaviour, class distribution, windows, and business-linked interpretation for both cities.	Mostly descriptive outputs with little interpretation or unequal treatment of the two cities.
Feature engineering & target construction	15%	Correct target, warm-up-only thresholds, leakage-free temporal features, and justified representation comparison.	Leakage, future-derived thresholds, weak temporal alignment, or unjustified transformations.
Correct Data Stream Mining methodology	20%	Fair prequential protocol, five mandatory models, same histories, correct metrics, and computational evidence.	Random splitting, learning before evaluation, inconsistent histories, missing mandatory models, or incorrect metrics.
Results interpretation & recommendations	15%	Connects data characteristics, target distribution, temporal behaviour, predictive metrics, computational cost, and operational meaning.	Reports the highest score as the conclusion without explaining event detection, cost, limitations, or business meaning.
Reproducibility, code quality & submission organisation	10%	Auditable code, clear README, versions/configuration recorded, outputs traceable to the submitted experiment.	Missing instructions/configuration, disorganised files, or results that cannot be traced to the submitted code.

CHAPTER 18

Critical Methodological Errors

These errors may substantially reduce the corresponding assessment component

- target leakage;
- using the target-defining variable at time t as an input at time t;
- estimating thresholds from future evaluation data;
- random train/test splitting instead of chronological replay;
- learning before evaluating the current event;
- different evaluation histories across models being compared;

- undocumented deletion or manipulation of observations;
- non-reproducible reported results.

A methodological error is not repaired by a high predictive score. If the experimental protocol is invalid, the resulting performance evidence cannot be interpreted as intended.

CHAPTER 19

Assignment Cards PA01–PA50

Each card is unique. The pair must use the assigned cities, business context, target, and business question unless the instructor explicitly approves a change. Across the catalogue, all 50 city pairs are unique and each of the 25 cities appears four times.

Registration

Record your pair and PA code in Moodle before beginning the final experiment. Your PA code is part of the submission filename and audit package.

Code / Target	Cities / Business Context	Business Question
PA01 T1	**Lisboa – Porto** Urban Walking Tour Operator	Which city offers more predictable conditions for scheduling walking tours with fewer rain interruptions?
PA02 T2	**Porto – Braga** Fresh-Food Cold Distribution Hub	Which location experiences fewer heat-stress periods that may increase cooling requirements?
PA03 T3	**Braga – Guimarães** Ornamental Plant Nursery	Which city presents fewer cold-stress periods for temperature-sensitive plants?
PA04 T4	**Guimarães – Viana do Castelo** Textile Storage Facility	How do humid and dry regimes differ between candidate warehouse locations?
PA05 T5	**Viana do Castelo – Vila Real** Construction Crane Operator	Which location presents more frequent strong-wind interruptions for lifting operations?
PA06 T6	**Vila Real – Bragança** Mountain Adventure Company	Which city presents more predictable weather windows for outdoor activities?
PA07 T1	**Bragança – Aveiro** Last-Mile Parcel Delivery	Which city exposes delivery operations to more frequent rain-related disruption?
PA08 T2	**Aveiro – Viseu** Chocolate Distribution Centre	Which location may require more cooling effort during relatively hot periods?
PA09 T3	**Viseu – Guarda** Greenhouse Heating Operator	Which location presents more frequent cold-stress conditions?
PA10 T4	**Guarda – Coimbra** Document and Archive Storage	Which city's external humidity regime could create greater environmental-control demand?
PA11 T5	**Coimbra – Leiria** Drone Mapping Company	Which city offers more stable wind conditions for routine aerial surveys?

Code / Target	Cities / Business Context	Business Question
PA12 **T6**	**Leiria – Castelo Branco** Open-Air Music Festival Producer	Which location offers more reliable operating windows for outdoor festivals?
PA13 **T1**	**Castelo Branco – Santarém** Weekly Open-Air Market Operator	Which city provides fewer rain disruptions for recurring outdoor markets?
PA14 **T2**	**Santarém – Setúbal** Fresh-Produce Distribution Warehouse	Which location experiences fewer high-temperature stress periods?
PA15 **T3**	**Setúbal – Évora** Vineyard Field-Operations Contractor	How does cold-stress frequency differ between the two operating regions?
PA16 **T4**	**Évora – Beja** Grain and Feed Storage Company	How different are the humidity regimes relevant to storage-environment control?
PA17 **T5**	**Beja – Faro** Outdoor Advertising Installer	Which city presents fewer strong-wind periods affecting installation work?
PA18 **T6**	**Faro – Portimão** Beachfront Hospitality Operator	Which location provides more reliable conditions for outdoor customer areas?
PA19 **T1**	**Portimão – Lagos** Camping and Outdoor Accommodation Company	Which city experiences fewer rain interruptions for outdoor accommodation activities?
PA20 **T2**	**Lagos – Funchal** Temperature-Sensitive Tourism Supply Company	Which location experiences fewer periods of relatively high temperature?
PA21 **T3**	**Funchal – Ponta Delgada** Outdoor Botanical Attraction	Which location experiences fewer relatively cold operating periods?
PA22 **T4**	**Ponta Delgada – Angra do Heroísmo** Electronics Equipment Storage & Rental	How do humidity regimes differ between the two island locations?
PA23 **T5**	**Angra do Heroísmo – Covilhã** Telecommunications Tower Maintenance	Which city provides more predictable wind conditions for scheduled maintenance?
PA24 **T6**	**Covilhã – Chaves** Outdoor Wedding & Events Venue	Which location offers more reliable weather windows for outdoor events?
PA25 **T1**	**Chaves – Lisboa** Road Maintenance Contractor	Which city has fewer rain-related interruptions for scheduled outdoor roadwork?
PA26 **T2**	**Lisboa – Vila Real** Temperature-Sensitive Food Processing Facility	Which candidate location experiences fewer heat-stress periods?
PA27 **T3**	**Porto – Bragança** Winter Road-Service Operator	Which city experiences a greater frequency of cold-stress periods relevant to winter operations?
PA28 **T4**	**Braga – Aveiro** Textile and Apparel Warehouse	Which city shows a more stable humidity regime for warehouse environmental management?
PA29 **T5**	**Guimarães – Viseu** Rooftop Solar Inspection Company	Which city provides more operating hours without strong-wind restrictions?
PA30 **T6**	**Viana do Castelo – Guarda** Seasonal Outdoor Fair Operator	Which city offers more predictable weather windows for seasonal fairs?

Code / Target	Cities / Business Context	Business Question
PA31 **T1**	**Vila Real – Coimbra** University Outdoor Events Service	Which campus location would experience fewer rain-related event disruptions?
PA32 **T2**	**Bragança – Leiria** Beverage Distribution Warehouse	Which location experiences fewer high-temperature periods relevant to cooling demand?
PA33 **T3**	**Aveiro – Castelo Branco** Outdoor Construction Crew	Which city presents fewer cold-stress periods for early outdoor operations?
PA34 **T4**	**Viseu – Santarém** Wooden Furniture Storage Company	How different are the humidity regimes between potential storage locations?
PA35 **T5**	**Guarda – Setúbal** Temporary Outdoor Structures Company	Which city presents fewer strong-wind periods affecting temporary structures?
PA36 **T6**	**Coimbra – Évora** Open-Air Cinema Operator	Which city offers more reliable evenings for recurring outdoor screenings?
PA37 **T1**	**Leiria – Beja** Last-Mile Grocery Delivery Fleet	Which city faces more frequent rain disruption for delivery operations?
PA38 **T2**	**Castelo Branco – Faro** Controlled Greenhouse Operator	Which location experiences fewer periods requiring increased cooling intervention?
PA39 **T3**	**Santarém – Portimão** Cycling Tourism Company	Which city offers fewer relatively cold periods for scheduled cycling activities?
PA40 **T4**	**Setúbal – Lagos** Bakery Ingredients Storage Facility	Which city has a humidity regime requiring greater environmental control?
PA41 **T5**	**Évora – Funchal** Wind-Turbine Maintenance Service	Which location provides more maintenance windows without strong-wind conditions?
PA42 **T6**	**Beja – Ponta Delgada** Outdoor Logistics Staging Area	Which location experiences fewer rain-or-wind interruptions to outdoor handling?
PA43 **T1**	**Faro – Angra do Heroísmo** Outdoor Excursion Operator	Which city provides more reliable rain-free operating periods for excursions?
PA44 **T2**	**Portimão – Covilhã** Outdoor Workforce Scheduling Service	Which location presents fewer high-temperature stress periods for outdoor workers?
PA45 **T3**	**Lagos – Chaves** Agricultural Plant Nursery	Which city presents fewer cold-stress periods for sensitive plant operations?
PA46 **T4**	**Funchal – Lisboa** Photography and Audiovisual Equipment Storage	Which city has the more favourable humidity regime from an environmental-control perspective?
PA47 **T5**	**Ponta Delgada – Porto** Port Infrastructure Maintenance Contractor	Which location offers more predictable low-wind maintenance windows?
PA48 **T6**	**Angra do Heroísmo – Braga** Outdoor Running Event Organiser	Which city offers more reliable weather windows for recurring outdoor races?
PA49 **T1**	**Covilhã – Guimarães** Outdoor Sports Academy	Which city experiences fewer rain interruptions during scheduled outdoor training?
PA50 **T2**	**Chaves – Viana do Castelo** Temperature-Sensitive Cosmetics Warehouse	Which location experiences fewer high-temperature periods relevant to cooling requirements?

Code / Target	Cities / Business Context	Business Question

The business question tells you what operational decision to support. The target tells you what event the streaming system must predict. Your report must first characterise the two city streams and the event itself, then evaluate the models, and finally produce separate business and technical recommendations.

CHAPTER 20

Final Submission Checklist

Before uploading your files to Moodle, verify every item below. The checklist is a final control; it does not replace the detailed requirements in the handbook.

- ☐ Pair and **PA code registered** and confirmed in Moodle.
- ☐ Both students can explain the complete project and their primary city analysis.
- ☐ The correct city pair, business context, and official target are used.
- ☐ All data-derived **thresholds estimated only from warm-up** observations and frozen before evaluation.
- ☐ There is **no target leakage**; current target-defining values are not current predictors.
- ☐ Chronological replay is preserved; no random train/test split is used.
- ☐ The **PREDICT → EVALUATE → LEARN** order is preserved.
- ☐ Both required feature representations are evaluated.
- ☐ All five mandatory models are evaluated under a fair common protocol.
- ☐ Target class distribution is inspected for both cities.
- ☐ Accuracy, Macro-F1, Precision, Recall, confusion matrix, prediction cost, and update cost are reported.
- ☐ At least **two original analyses or visualisations** beyond starter-code outputs are included.
- ☐ Business Recommendation and Technical Recommendation are clearly separated.
- ☐ Limitations are stated and conclusions do not exceed the evidence available.
- ☐ The mandatory **AI Assistance Statement** is included and truthful.
- ☐ `PAxx_GroupYY_Report.pdf` and `PAxx_GroupYY_Audit.zip` are named correctly.
- ☐ Code, configurations, data instructions, and main results are reproducible and auditable.

Final reminder

Your task is not to find the model with the highest score.

Your task is to understand a real stream, transform it into a meaningful prediction problem, evaluate incremental learners correctly, and use evidence to support an operational decision.